CASE STUDY REPORT

NimbusEdge Technologies Network Outage Incident

Routing Protocol Failures – March 2024

Executive Summary

NimbusEdge Technologies Pvt. Ltd. experienced a cascading network outage in March 2024 that disrupted enterprise client services for approximately nine hours. The incident was not caused by one isolated fault. Four routing-related problems occurred across different network segments: a RIP hop-count limitation at Chennai HQ, an IGRP convergence problem on the Bengaluru–Pune regional link, an OSPF adjacency failure at Hyderabad caused by an incorrect Area ID, and a BGP failover-policy problem affecting the dual-ISP connection. Together, these failures demonstrate the importance of correct routing design, configuration validation, monitoring, convergence testing and disaster-recovery procedures.

Central lesson: Reliable routing requires more than enabling a protocol. Its limits, metrics, neighbor requirements, policies and failure behaviour must be understood, configured correctly, monitored continuously and tested under realistic failure conditions.

Q1. RIP – Chennai HQ Campus

a) Understanding of the Case (25 Marks)

In March 2024, the Delivery department subnet at the Chennai headquarters was moved to a new subnet. The new addressing and routing arrangement caused the destination to become 16 router hops away from part of the RIP-based campus network. This is critical because RIP uses hop count as its routing metric and has a maximum usable hop count of 15. A metric of 16 is treated as infinity, meaning the destination is considered unreachable. Therefore, the configuration change was the direct technical trigger for the failure.

Incident sequence

First, the Delivery subnet was changed. Second, the new route required more than the maximum RIP distance. Third, RIP treated the destination as unreachable. Fourth, there was no effective automatic alert for the loss of reachability. Finally, support tickets became the practical indication that the subnet was unavailable. The incident therefore combined a routing-protocol limitation with a monitoring gap.

b) Application of Concepts (25 Marks)

RIP is a distance-vector routing protocol. Its main routing metric is hop count, representing the number of routers crossed to reach a destination. RIP accepts a maximum of 15 hops. A route at 16 hops is considered unreachable. Thus, 1–15 hops are reachable, while 16 hops means unreachable. Because the new Delivery subnet was effectively 16 hops away, RIP could not maintain it as a valid destination. This explains why the route disappeared instead of simply becoming a slower route.

c) Analysis (20 Marks)

The absence of an automatic alert is an important operational weakness. Routing protocols make path decisions, but monitoring systems must also verify whether important networks are actually reachable. A network can have functioning router interfaces while a particular destination is unavailable. This incident shows why monitoring should include end-to-end reachability. It also demonstrates the limitation of a distance-vector protocol with a hard 15-hop boundary when a network grows or addressing changes.

d) Solution Design (20 Marks)

The immediate solution is to review the campus topology and addressing scheme so critical destinations do not exceed RIP's hop-count limitation. Unnecessary routing hops should be reduced where practical. The stronger long-term solution is migration to OSPF. OSPF uses a cost-based link-state approach, has no RIP-style 15-hop limitation, converges more effectively, and is more suitable for a growing enterprise network. Migration should include route testing, monitoring validation and a rollback plan.

e) Clarity (10 Marks)

Before: Delivery subnet → multiple routers → 16 hops → RIP metric 16 → destination unreachable.
After: Delivery subnet → optimized topology / OSPF → valid route → normal connectivity.
Recommendation: Replace RIP with OSPF for the growing campus network and add monitoring for important subnet reachability.

Q2. IGRP – Bengaluru–Pune–Hyderabad Regional Link

a) Understanding of the Case (25 Marks)

The Bengaluru–Pune regional connection experienced an ISP-side fault. The direct link degraded to about 2 Mbps, making it unsuitable for normal enterprise traffic. A healthier alternate path was available through Hyderabad: Bengaluru → Hyderabad → Pune. However, the IGRP-based routing system did not move traffic to the alternate path quickly enough. Client applications therefore experienced severe latency while the degraded path remained in use.

b) Application of Concepts (25 Marks)

IGRP uses a composite routing metric based mainly on bandwidth and delay. Using the simplified default form requested for this case, the bandwidth component is 10,000,000 divided by the minimum bandwidth in kbps. For a 2 Mbps link: 2 Mbps = 2000 kbps; bandwidth component = 10,000,000 / 2000 = 5000. The degraded bandwidth therefore produces a much less attractive metric. After recalculation and convergence, the healthier alternate route should become preferable.

c) Analysis (20 Marks)

A physical link can degrade before a routing protocol has completely converged. The sequence is: physical condition changes → router detects the change → routing information is updated → neighboring routers receive the information → metrics are recalculated → the best route is installed. The time between the real network change and the final routing decision is convergence time. This incident therefore represents a convergence-speed problem as well as a metric-recalculation issue.

d) Solution Design (20 Marks)

NimbusEdge can temporarily tune IGRP parameters where appropriate, but the preferred strategic solution is migration to OSPF. OSPF maintains link-state information and generally provides faster, more predictable convergence and better scalability. The migration should test degraded-link scenarios, route preference, convergence time and application performance. The alternate Bengaluru → Hyderabad → Pune path must be verified as the expected backup before production use.

e) Clarity (10 Marks)

Normal condition: Bengaluru → Pune direct path is preferred.
Failure condition: Bengaluru → Pune becomes a 2 Mbps degraded path.
Expected recovery: Bengaluru → Hyderabad → Pune becomes the preferred healthy path.
Recommendation: move the regional segment toward OSPF and validate failover under simulated ISP degradation.

Q3. OSPF – Core Backbone Migration

a) Understanding of the Case (25 Marks)

During NimbusEdge's migration to multi-area OSPF, the Hyderabad router was configured with an incorrect Area ID. OSPF uses areas to create a hierarchical link-state architecture, and routers on a common OSPF segment must agree on the appropriate area. Because the Area ID did not match the expected configuration, the Hyderabad router failed to form the required neighbor adjacency. Routing information was not exchanged as intended, producing a routing black hole for affected traffic.

b) Application of Concepts (25 Marks)

OSPF neighbor formation depends on compatible parameters such as Area ID, Hello interval, Dead interval, authentication settings where used, and network/subnet configuration. In this case, the key mismatch was the Area ID. For example, if one side uses Area 0 while the corresponding side uses Area 1, the expected adjacency cannot form across the link. Without the adjacency, the routers cannot exchange the expected link-state information over that relationship.

c) Analysis (20 Marks)

This failure can be harder to identify than a physical link failure. A physical failure normally produces an interface-down event that can be detected quickly. A configuration error can leave the physical interface up while the routing adjacency remains unavailable. The RCA therefore points to a process problem as well as a technical problem: the configuration change was allowed into production without sufficient pre-change verification and post-change validation. This shows why change management is part of network reliability.

d) Solution Design (20 Marks)

The technical fix is to correct the Area ID and verify the configuration at both ends. The process fix should be equally strong. NimbusEdge should require peer review, pre-change topology checks, explicit verification of Area IDs, Hello/Dead timers and authentication, post-change neighbor verification, routing-table checks, ping/traceroute tests and a documented rollback plan. These checks should be recorded in the change ticket before the change is closed.

e) Clarity (10 Marks)

Before: Hyderabad router → incorrect Area ID → OSPF adjacency failure → incomplete route exchange → routing black hole.
After: Hyderabad router → correct Area ID → adjacency established → link-state information exchanged → correct routing.
Multi-area OSPF is appropriate because it provides hierarchical scalability, faster convergence and better fault isolation.

Q4. BGP – ISP Peering and Failover

a) Understanding of the Case (25 Marks)

NimbusEdge used two ISP connections, with ISP-A as the primary and ISP-B as the backup. When ISP-A failed, the expected behaviour was for BGP to withdraw the unavailable path and select ISP-B. Instead, the backup path was not selected automatically as expected. Manual intervention was required, extending the outage and contributing to a formal client SLA breach. The incident shows that redundancy is useful only when the failover policy itself is correctly configured and tested.

b) Application of Concepts (25 Marks)

BGP selects routes using path attributes and policy. Important attributes for this case include LOCAL_PREF, AS-PATH and MED. LOCAL_PREF is commonly used within an autonomous system to express outbound path preference; a higher value is normally preferred. AS-PATH provides the sequence of autonomous systems a route has traversed, and a shorter path can be preferred when other policy conditions are equal. MED can influence which external entry point is preferred between neighboring autonomous systems. An incorrect policy can therefore prevent the backup ISP from becoming the selected path when the primary route fails.

c) Analysis (20 Marks)

The key operational problem was insufficient failover-policy testing. A BGP configuration can appear correct during normal operation because the primary ISP is available. The real test occurs when the primary path disappears. The business consequence chain was: ISP-A failure → expected failover did not work → manual action required → longer connectivity disruption → client services affected → SLA commitments impacted. Therefore, BGP policy validation and failure simulation are essential.

d) Solution Design (20 Marks)

NimbusEdge should configure a clear preference hierarchy. During normal operation, ISP-A should have the preferred policy, while ISP-B should remain a valid lower-preference backup. When ISP-A becomes unavailable, its routes should be withdrawn or become unusable, allowing ISP-B to become the best available path. The organization should validate LOCAL_PREF and other relevant policy attributes, monitor BGP session state, and periodically test actual ISP failure scenarios. Every change should have peer review and rollback steps.

e) Clarity (10 Marks)

Before: ISP-A → preferred route; ISP-B → backup route; ISP-A fails → incorrect policy → backup not selected automatically.
After: ISP-A → preferred route; ISP-B → backup route; ISP-A fails → primary withdrawn → ISP-B becomes best available route.
Why BGP? BGP is appropriate for dual-ISP external routing because it is designed for inter-autonomous-system routing and provides policy controls for primary/backup selection.

Comparative Summary

Protocol	Root Cause	Impact	Recommended Solution
RIP	New subnet exceeded 15-hop limit	Delivery subnet unreachable	Optimize topology and migrate to OSPF
IGRP	Slow response to degraded ISP path	High latency	Improve convergence / migrate to OSPF
OSPF	Incorrect Area ID	Adjacency failure / black hole	Correct Area ID + change validation
BGP	Incorrect failover policy	Backup ISP not selected	Correct policy + failover testing

Overall Root-Cause Analysis

The outage was a cascading incident rather than a single protocol failure. The RIP problem came from a hard protocol limitation combined with insufficient reachability monitoring. The IGRP problem involved real-time link degradation and delayed convergence toward an alternate path. The OSPF problem was introduced during a planned migration because an incorrect Area ID was not detected before production deployment. The BGP problem was a policy and failover-validation failure in a dual-ISP design. These cases represent four major routing risks: protocol limitations, convergence delay, configuration mismatch and policy failure.

Recommended Enterprise Network Architecture

NimbusEdge should standardize internal routing around OSPF, preferably using a structured multi-area design as the network grows. OSPF should handle campus, regional and core internal routing. BGP should remain at the Internet edge for dual-ISP connectivity and policy-based external routing. RIP should be removed from critical production segments where practical, and IGRP should not be relied upon for new regional deployments.

Operational Improvements

1. Mandatory peer review for routing changes.
2. Pre-change configuration and topology validation.
3. Monitoring of both interface health and end-to-end reachability.
4. OSPF neighbor and BGP session verification after routing changes.
5. Scheduled ISP failover and routing-convergence tests.
6. Documented rollback procedures for major changes.
7. Change records linking configuration changes to incidents.
8. Regular simulation of real failure conditions.

Final Conclusion

The NimbusEdge incident demonstrates that routing reliability depends on appropriate protocol selection, correct configuration, effective monitoring and realistic failure testing. RIP failed because the destination exceeded its hop-count limit; IGRP did not react quickly enough to the degraded regional link; OSPF lost an adjacency because of an incorrect Area ID; and BGP failed to provide the expected automatic ISP failover because of policy configuration. The recommended approach is to use OSPF for internal routing, BGP for dual-ISP connectivity, and strong change-management and monitoring practices throughout the network. This will make NimbusEdge's infrastructure more scalable, resilient and easier to operate.